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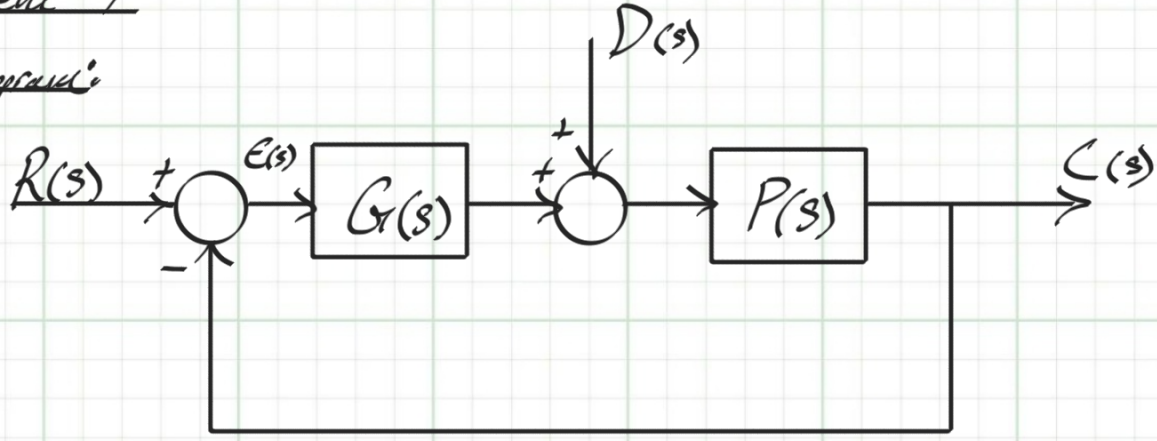
Problem #1

Diagram:



Given:

* $G(s) = S$
* $P(s) = \frac{7}{s+2}$

Determine

a) Calculate Steady State error $e_{ss}(t)$ | $t \to \infty$, $R(s) = \frac{3}{s}$, $D(s) = 0$

b) Verify results from a) using simulink

c) Calculate Steady State error $e_{ss}(t)$ | $t \to \infty$, $R(s) = 0$, $D(s) = -\frac{1}{s}$

d) Verify results from c) using simulink

e) Calculate Steady State error $e_{ss}(t)$ | $t \to \infty$, $R(s) = \frac{3}{s}$, $D(s) = -\frac{1}{s}$

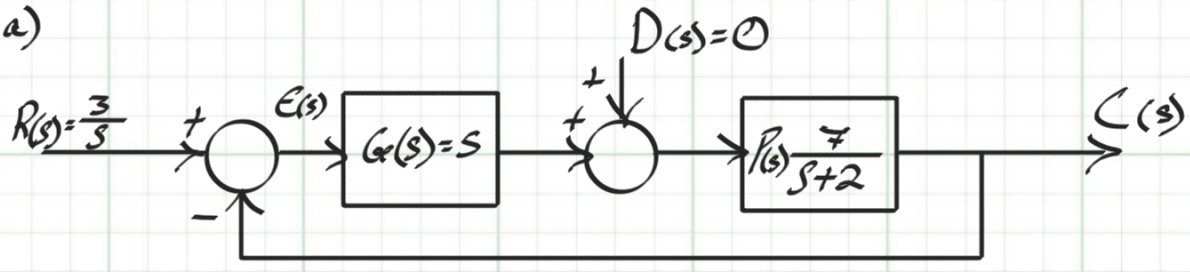
f) Verify results from e) using simulink

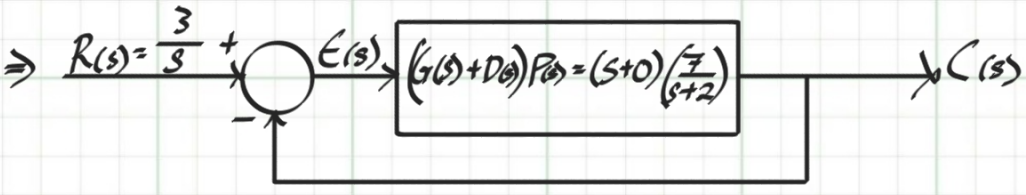
Note:

* S.S. error is relative to R(s)

Analysis:

a)





$$E(s) = R(s) - C(s) \quad | \quad C(s) = \left(\frac{(G(s) + D(s))P(s)}{1 + (G(s) + D(s))P(s)} \right) R(s)$$

$$\Rightarrow E(s) = \frac{R(s)}{1 + (G(s)+D(s))P(s)} = \frac{\frac{3}{s}}{1 + 5\frac{7}{s+2}} = \frac{\frac{3}{s}}{\frac{s+2+35}{s+2}} = \frac{3(s+2)}{s^2+37s}$$

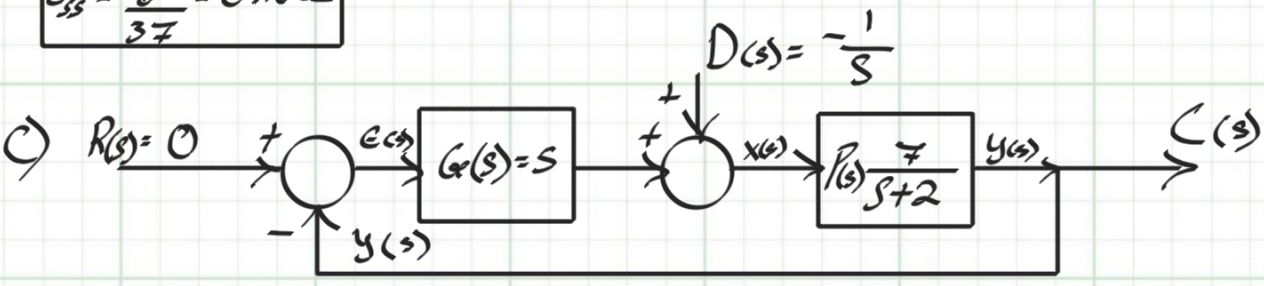
$$E(s) = \frac{3(s+2)}{s(s+37)}$$

$$e(t) @ (t=\infty) = e_{ss}$$

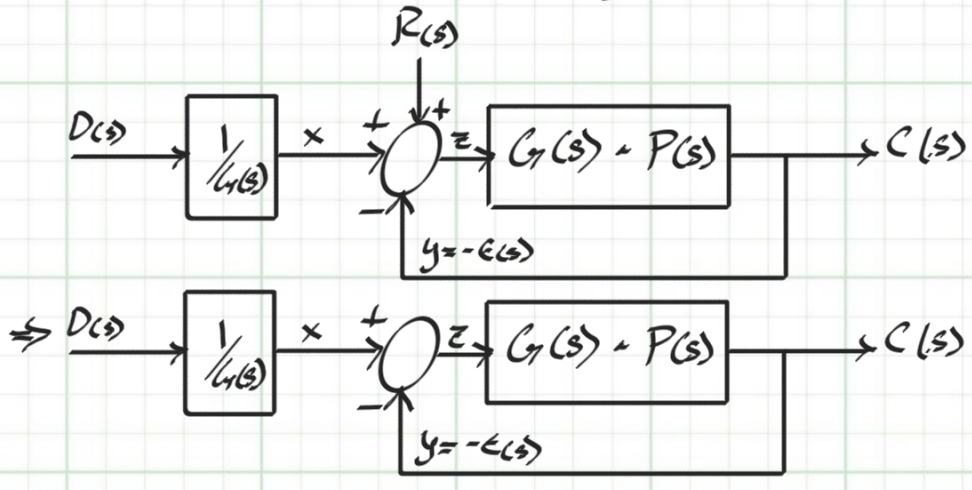
$$\Rightarrow e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\,\frac{3(s+2)}{s(s+37)} = \frac{6}{37} = 0.1622$$

$$\boxed{e_{ss} = \frac{6}{37} = 0.1622}$$

c) $R(s) = 0$



$\because E(s) = R(s)^{0} - y(s) \quad \therefore E(s) = -y(s) = -C(s)$



$\Rightarrow$

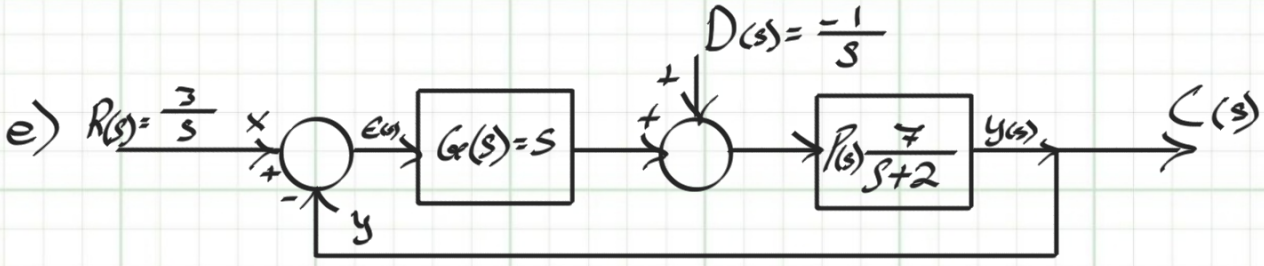
$$\Rightarrow C(s) = \frac{D(s)}{G(s)} \cdot \frac{G(s)P(s)}{1+G(s)P(s)} = \frac{D(s)P(s)}{1+G(s)P(s)} = y = -E(s)$$

$$= \frac{\frac{-1}{s}\left(\frac{7}{s+2}\right)}{1+\frac{35}{s+2}} \Rightarrow \frac{-7/s}{s+37} = \frac{-7}{s(s+37)}$$

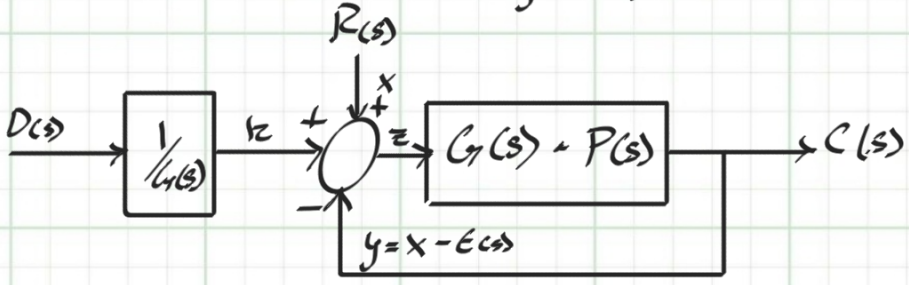
$$E(s) = \frac{7}{s(s+37)}$$

$$e(t) @ (t=\infty) = e_{ss} = \lim_{s\to 0} sE(s) = \frac{7}{37} = 0.1892$$

$$\boxed{e_{ss} = \frac{7}{37} = 0.1892}$$

e) 

$* \epsilon(s) = x - y \;\Big|\; x = 3/s,\ y = C(s)$



$$C(s) = \left(R(s) + \frac{D(s)}{G_c(s)}\right)\left(\frac{G_c(s)P(s)}{1 + G_c(s)P(s)}\right)$$

$$= \left(R(s)G_c(s) + D(s)\right)\frac{P(s)}{1 + G_c(s)P(s)}$$

$$= \frac{P(s)\left(R(s)G_c(s) + D(s)\right)}{1 + G_c(s)P(s)}$$

$$= \frac{7(13s - 1)}{(s+2)(s)} \times \left(\frac{(s+2)}{s+37}\right) = \frac{98}{s(s+37)} = C(s) = y$$

$$\Rightarrow \epsilon(s) = \frac{98}{s(s+37)} - \frac{3}{s} \Rightarrow \frac{98 - 3(s+37)}{(s+37)s} = \frac{-3s - 13}{s(s+37)}$$

$$e(t)@(t=\infty) = e_{ss} = \lim_{s \to 0} s\epsilon(s) = s\left(\frac{-3s-13}{s(s+37)}\right) = \frac{-13}{37} = -0.3514$$

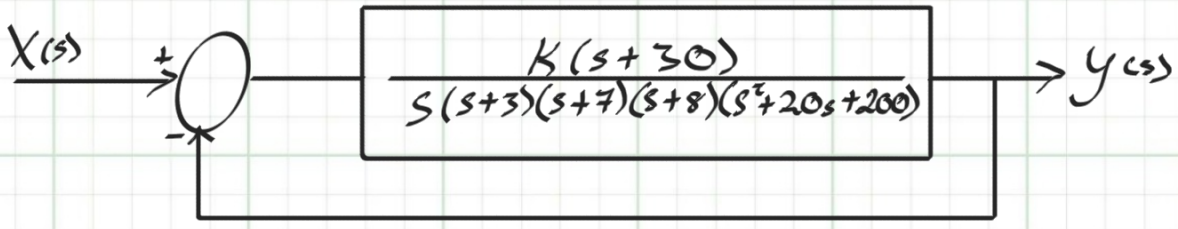
$$\boxed{e_{ss} = \frac{-13}{37} = -0.3514}$$

b), d), f) Verification results/outputs/simulink models shown in Apendix A

(Simulink models for each a), c), & e) validated hand calculated e_{ss} values).

Problem #2

Diagram:



For the following write Programs in Matlab

a) Display the root locus

b) Display close-up using axes limited to (-2, 2) on both real and imaginary axis

c) Overlay the 0.707 damping ratio line on the close-up root locus

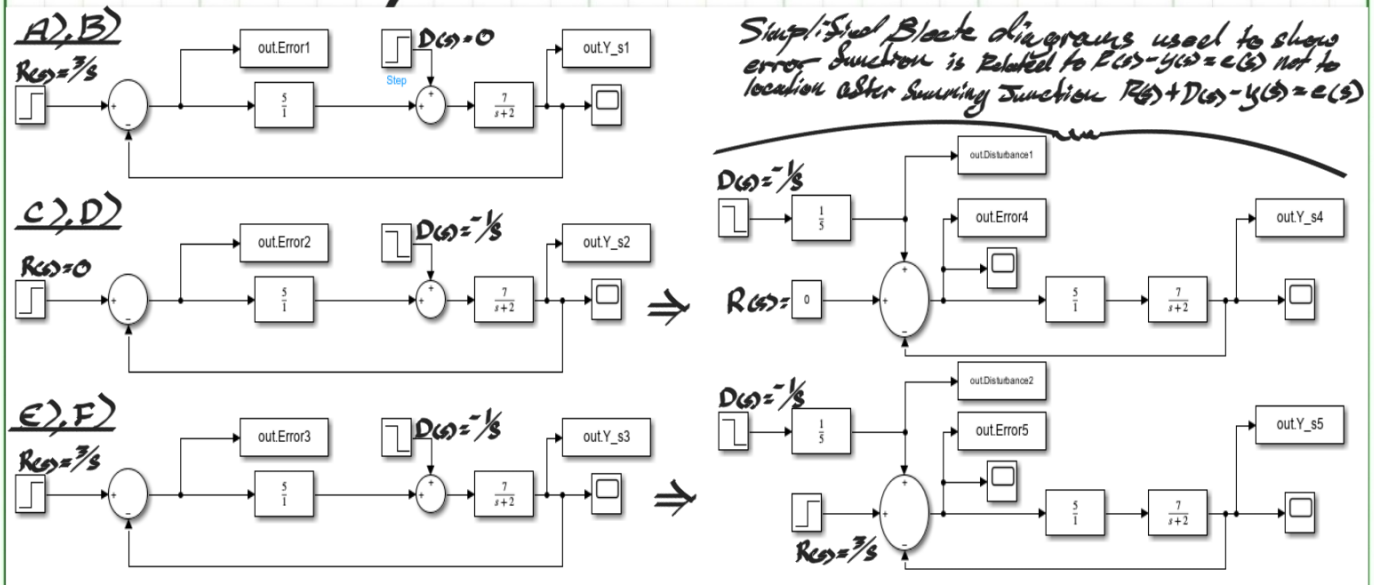
d) On plot interactively select point where the root locus crosses the damping ratio. Give the gain @ this point.

e) Generate the step at the gain for the 0.707 damping ratio.

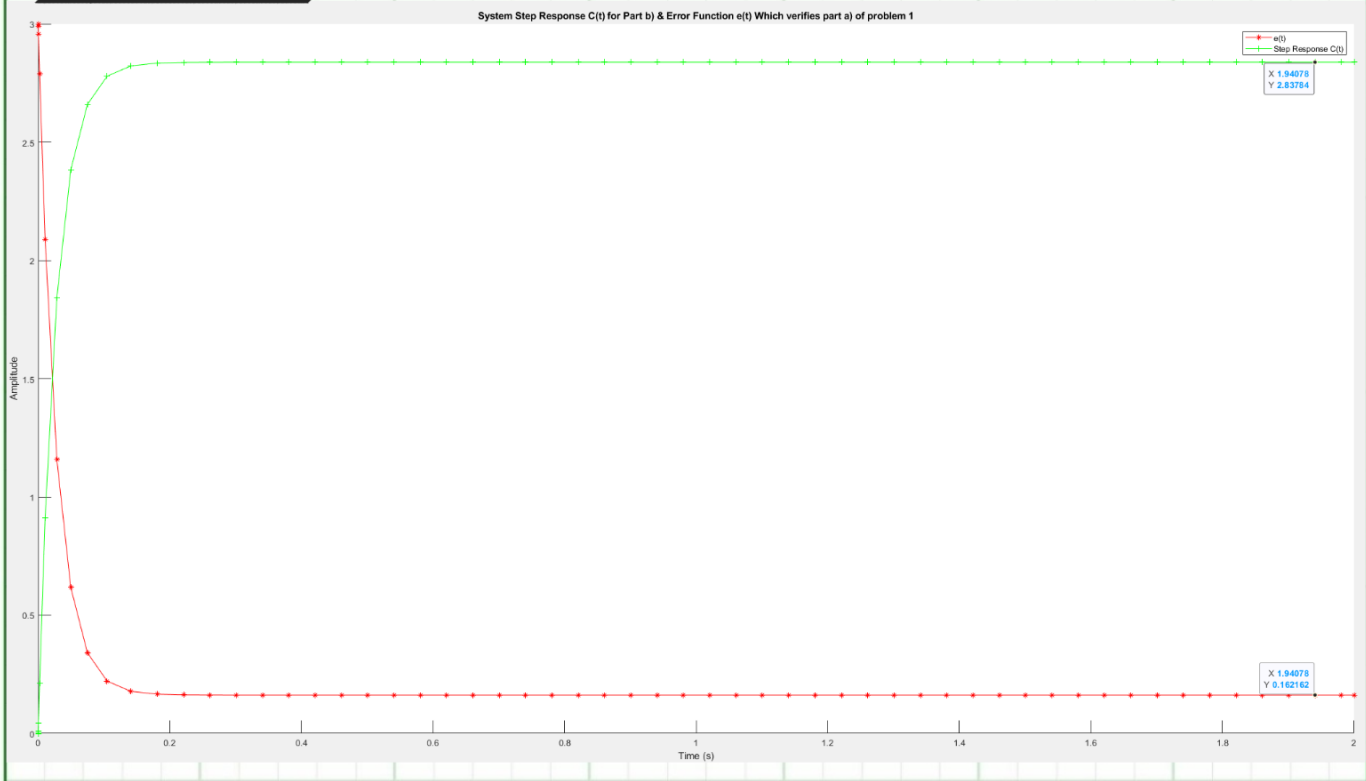
* The following provided in Apendix B

- printout of Matlab code
- root locus plot
- damping ratio lines
- step response

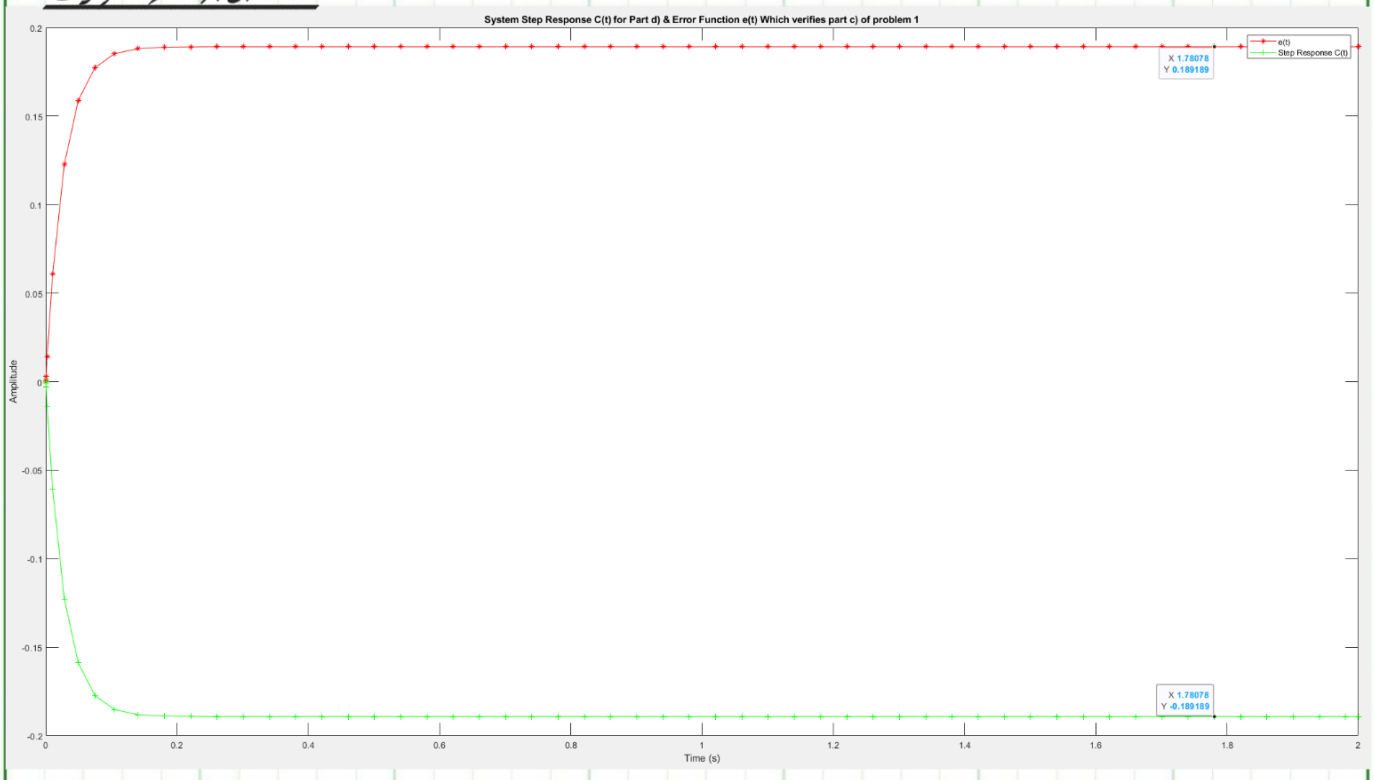
Appendix A



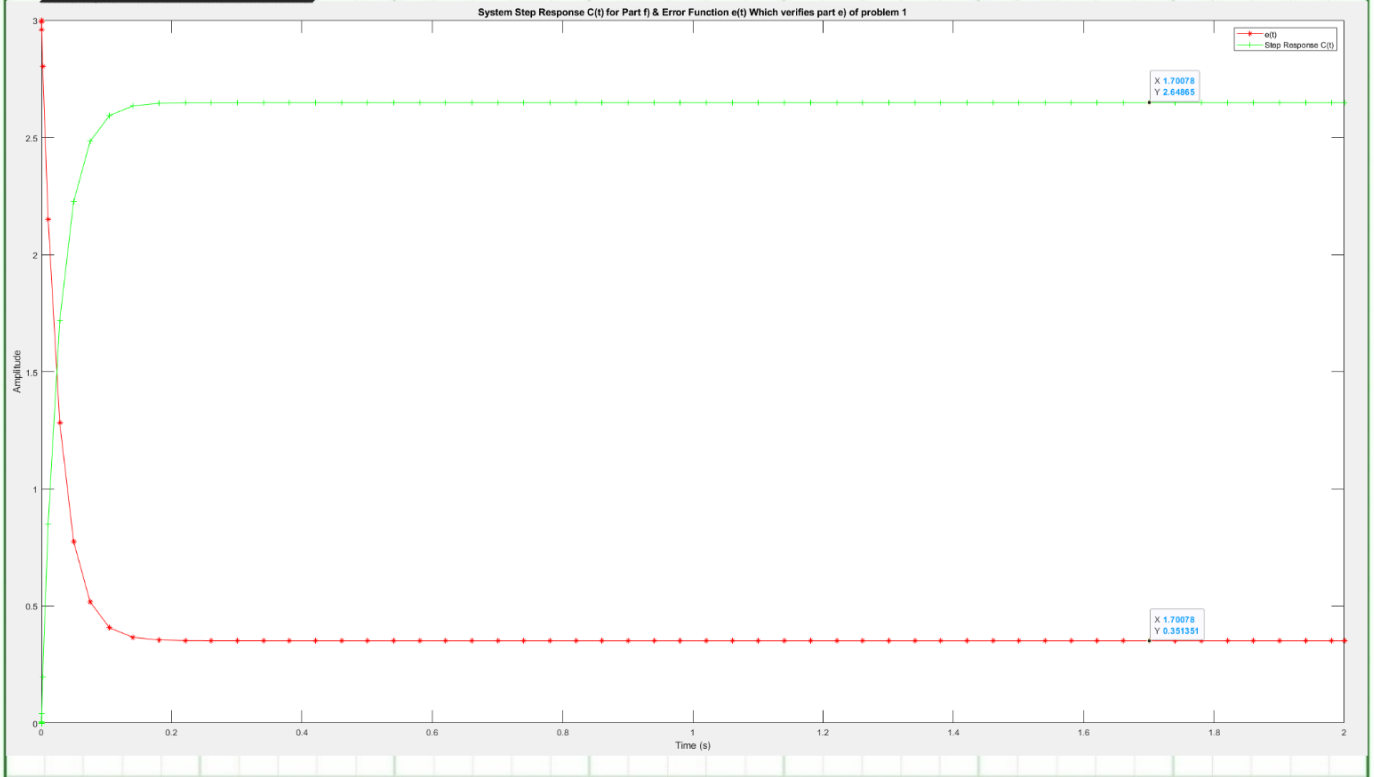
* A), B) Plot



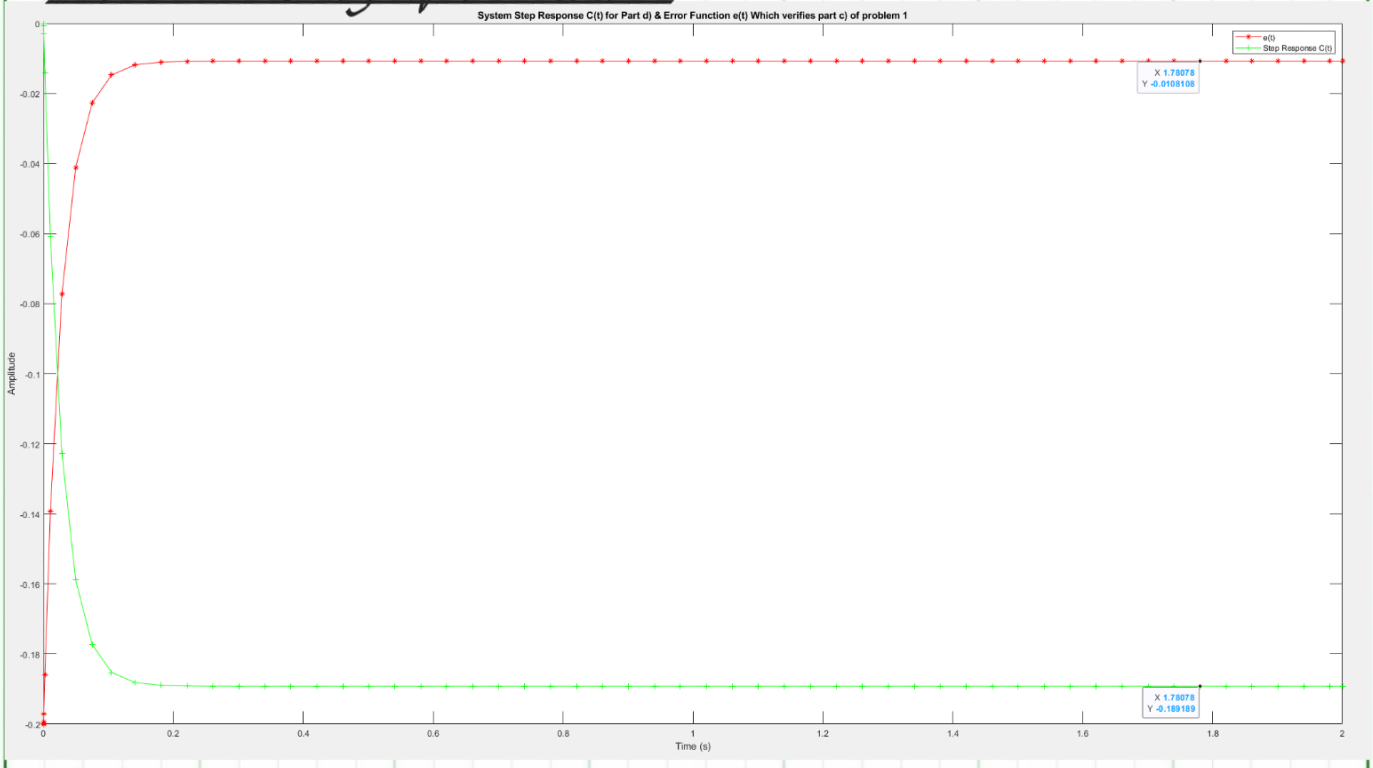
* c), d) Plot



* e), f) Plot



*C), D> Plot Using Simplified Model



- Notice the S.S error shows the error w/ respect to Both Rcss + Dcs this is not the correct error function w/ respect to Rcs> (shown for completeness and correlation to correct formulation as shown in (*C), D> Plot) figure above.

*E), F> Plot Using Simplified Model

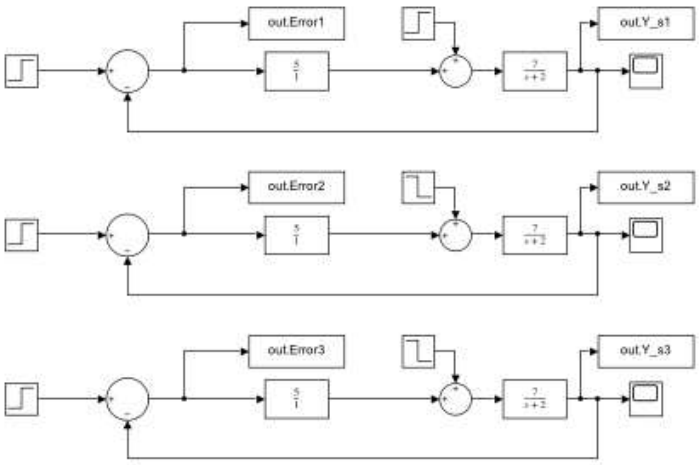
- Notice the S.S error shows the error w/ respect to Both Rcss + Dcs this is not the correct error function w/ respect to Rcs> (shown for completeness and correlation to correct formulation as shown in (*E), F> Plot) figure above.

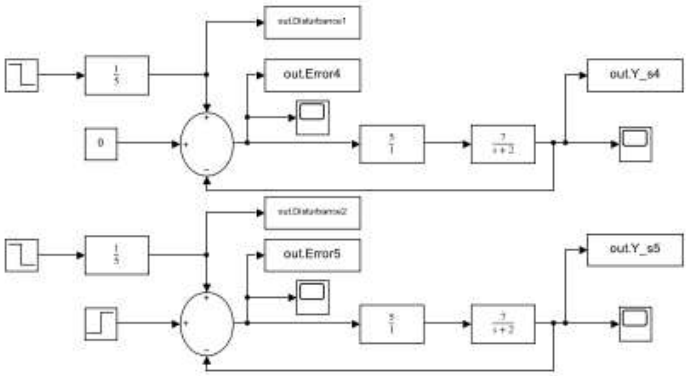
```
clc
clear
clf
%{
once block diagrams for a),c),and e) are properly set up in simulinks using
simout blocks at the respective location for error anlysis. Simout block
stores simulink data into the workspace.Save data into accessable file and
then load it back into manipulate.
%}

load("CA1-P1_Error_out_Data.mat")
Error_Data=[out.Error1.Data,out.Error2.Data,out.Error3.Data,out.Error4.Data,out.Error5.Data];
Error_time=out.Error1.Time;
Y_s_out=[out.Y_s1.Data,out.Y_s2.Data,out.Y_s3.Data,out.Y_s4.Data,out.Y_s5.Data];
Y_time=out.Y_s1.Time;
Dist_Data=[out.Disturbance1.Data,out.Disturbance2.Data];
%%
%plot the step response data w/ overlay of the error function for anlysis

hold on
problem_Part=["b)" "d)" "f)" "d)" "f)" "a)" "c)" "e)" "c)" "e)"];
for i=1:5
    title_text='System Step Response C(t) for Part '+problem_Part(i)+[' & ' ...
        'Error Function e(t) Which verifies part ']+problem_Part(5+i)+' of problem 1';
    figure(i)
    plot(Error_time, Error_Data(:,i),'r*-',Y_time,Y_s_out(:,i),'g+-')
    title(title_text)
    legend('e(t)','Step Response C(t)')
    xlabel('Time (s)')
    ylabel("Amplitude")
end
%%
% use Error data in comparision of Step Response data from the nonaltered simulink models to show the Steady
% State error

for i=1:3
 fprintf(['Given Model %s produces a Step Response C(t) with a final value of %f\n' ...
     'Since R(t->inf) reaches a Final Value of %f before input into system\n' ...
     'i.e R(t) @ t=0 (time scale for system)= %f\n' ...
     '  * Implies a Steady State Error e_(ss) = %f w/ respect to R(S) as the input step funtction.\n' ...
     '      e_(ss) = R(t = 0) - C(t -> inf) ---> (%f) - (%f) = %f\n\n'] ...
     ,problem_Part(i),Y_s_out(end,i),Error_Data(1,i),Error_Data(1,i),Error_Data(end,i), Error_Data(1,i), ...
     Y_s_out(end,i),Error_Data(end,i))
end
%%
% use Error data in comparision of Step Response data from the altered/ simplified simulink models to show the Steady
```

```
% State error in relations to R(s) and only R(s) because simplified models
% alters the linkage were the error is noticably input into the system.
% therefore this is to show that adjusting the models one needs to be aware
% of the data one is directly looking for.


for i=1:2
 fprintf(['Given Simplified Model %s  produces a Step Response C(t) with a ' ...
     'final value of %f and R(t = 0) = %f\n' ...
     'While the Disturbance D(s) into system after simplification produces ' ...
     'a constant value of x(t) = %f into system.\n' ...
     'Implies e_(ss) = R(t = 0) - C(t -> Inf) + x(t) = (%f) - (%f) + (%f)= %f\n\n' ...
     '  * This however is not the same error and is the error with respect ' ...
     'to Both R(s) and D(S) after the summing junction\n' ...
     '     The actual steady state error e_ss w/ respect to R(s) only is ' ...
     'Calculated as follows:\n' ...
     '          e_(ss) = R(t = 0) - C(t -> inf) = (%f) - (%f) = %f\n\n'] ...
     ,problem_Part(1+i),Y_s_out(end,i+3),Error_Data(1,i+1),Dist_Data(1,i),Error_Data(1,↙
i+1), ...
     Y_s_out(end,i+3),Dist_Data(1,i),Error_Data(end,i+3),Error_Data(1,i+1),Y_s_out(end,↙
i+3),Error_Data(end,i+1))
end
```

```
Given Model b) produces a Step Response C(t) with a final value of 2.837838
Since R(t->inf) reaches a Final Value of 3.000000 before input into system
i.e R(t) @ t=0 (time scale for system)= 3.000000
  * Implies a Steady State Error e_(ss) = 0.162162 w/ respect to R(S) as the input step↙
funtction.
      e_(ss) = R(t = 0) - C(t -> inf) ---> (3.000000) - (2.837838) = 0.162162

Given Model d) produces a Step Response C(t) with a final value of -0.189189
Since R(t->inf) reaches a Final Value of 0.000000 before input into system
i.e R(t) @ t=0 (time scale for system)= 0.000000
  * Implies a Steady State Error e_(ss) = 0.189189 w/ respect to R(S) as the input step↙
funtction.
      e_(ss) = R(t = 0) - C(t -> inf) ---> (0.000000) - (-0.189189) = 0.189189

Given Model f) produces a Step Response C(t) with a final value of 2.648649
Since R(t->inf) reaches a Final Value of 3.000000 before input into system
i.e R(t) @ t=0 (time scale for system)= 3.000000
  * Implies a Steady State Error e_(ss) = 0.351351 w/ respect to R(S) as the input step↙
funtction.
      e_(ss) = R(t = 0) - C(t -> inf) ---> (3.000000) - (2.648649) = 0.351351

Given Simplified Model d)  produces a Step Response C(t) with a final value of -0.189189↙
and R(t = 0) = 0.000000
While the Disturbance D(s) into system after simplification produces a constant value of↙
x(t) = -0.200000 into system.
Implies e_(ss) = R(t = 0) - C(t -> Inf) + x(t) = (0.000000) - (-0.189189) + (-0.200000)=↙
-0.010811

  * This however is not the same error and is the error with respect to Both R(s) and D↙
(S) after the summing junction
     The actual steady state error e_ss w/ respect to R(s) only is Calculated as follows:
          e_(ss) = R(t = 0) - C(t -> inf) = (0.000000) - (-0.189189) = 0.189189

Given Simplified Model f)  produces a Step Response C(t) with a final value of 2.648649↙
and R(t = 0) = 3.000000
While the Disturbance D(s) into system after simplification produces a constant value of↙
x(t) = -0.200000 into system.
Implies e_(ss) = R(t = 0) - C(t -> Inf) + x(t) = (3.000000) - (2.648649) + (-0.200000)=↙
0.151351

  * This however is not the same error and is the error with respect to Both R(s) and D↙
(S) after the summing junction
     The actual steady state error e_ss w/ respect to R(s) only is Calculated as follows:
          e_(ss) = R(t = 0) - C(t -> inf) = (3.000000) - (2.648649) = 0.351351

>>
```

Problem1_CA2





Jarel Shahim

15-Nov-2023 07:46:56

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Model - Problem1_CA2

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Full Model Hierarchy

1. Problem1_CA2

Simulation Parameter	Value
Solver	VariableStepAuto
RelTol	1e-3
Refine	1
MaxOrder	5
ZeroCross	on

[more info]

Machine - Problem1_CA2

[more info]

System - Problem1_CA2

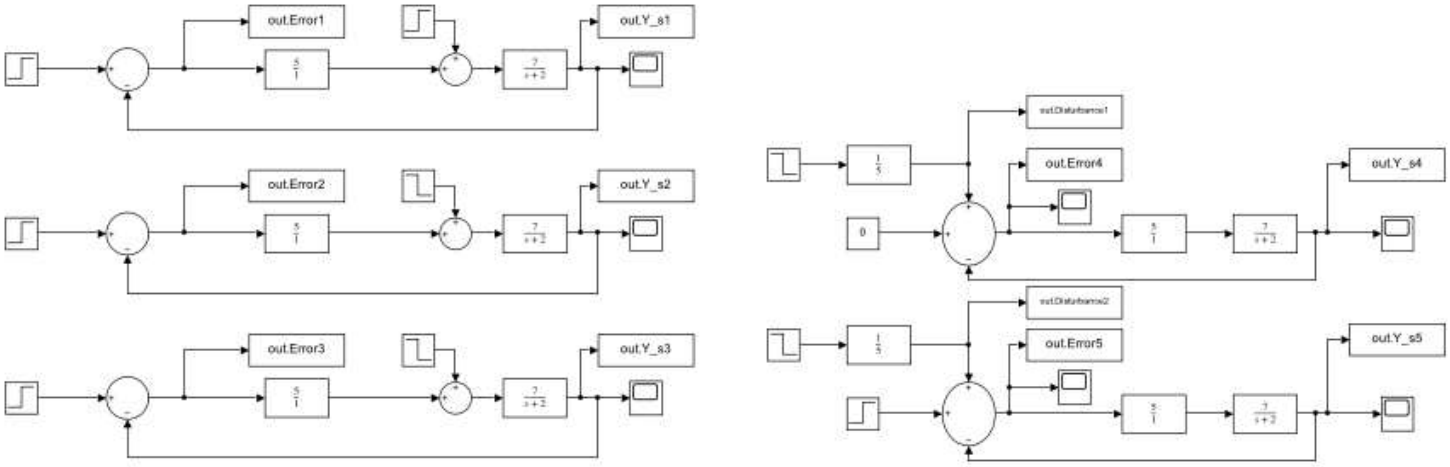


Table 1. Constant Block Properties

Name	Value	Out Data Type Str	Lock Scale	Sample Time	Frame Period
Constant1	0	Inherit: Inherit from 'Constant value'	off	inf	inf

Table 2. Step Block Properties

Name	Time	Before	After	Out Data Type Str	Sample Time	Zero Cross
Step	0	0	0	double	0	on
Step1	0	0	-1	double	0	on
Step2	0	0	3	double	0	on
Step3	0	0	-1	double	0	on
Step4	0	0	0	double	0	on
Step5	0	0	-1	double	0	on
Step6	0	0	3	double	0	on
Step7	0	0	-1	double	0	on
Step8	0	0	3	double	0	on

Table 3. Sum Block Properties

Name	Icon Shape	Inputs	Collapse Mode	Collapse Dim	Out Data Type Str	Accum Data Type Str	Input Same DT	Lock Scale	Rnd Meth	Saturate On Integer Overflow
Sum	round	++|	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum1	round	|+-	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off

Name	Icon Shape	Inputs	Collapse Mode	Collapse Dim	Out Data Type Str	Accum Data Type Str	Input Same DT	Lock Scale	Rnd Meth	Saturate On Integer Overflow
Sum2	round	++|	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum3	round	+||+||-	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum4	round	|+-	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum5	round	++|	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum6	round	|+-	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off
Sum7	round	+||+||-	All dimensions	1	Inherit: Inherit via internal rule	Inherit: Inherit via internal rule	off	off	Floor	off

Table 4. ToWorkspace Block Properties

Name	Variable Name	Max Data Points	Decimation	Save Format	Save 2DSignal	Fixpt As Fi
To Workspace	Error1	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace1	Y_s1	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace10	Disturbance1	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace11	Disturbance2	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace2	Error4	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace3	Y_s4	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace4	Error2	inf	1	Timeseries	3-D array (concatenate along third dimension)	on

Name	Variable Name	Max Data Points	Decimation	Save Format	Save 2DSignal	Fixpt As Fi
To Workspace5	Y_s2	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace6	Error3	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace7	Y_s3	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace8	Error5	inf	1	Timeseries	3-D array (concatenate along third dimension)	on
To Workspace9	Y_s5	inf	1	Timeseries	3-D array (concatenate along third dimension)	on

Table 5. TransferFcn Block Properties

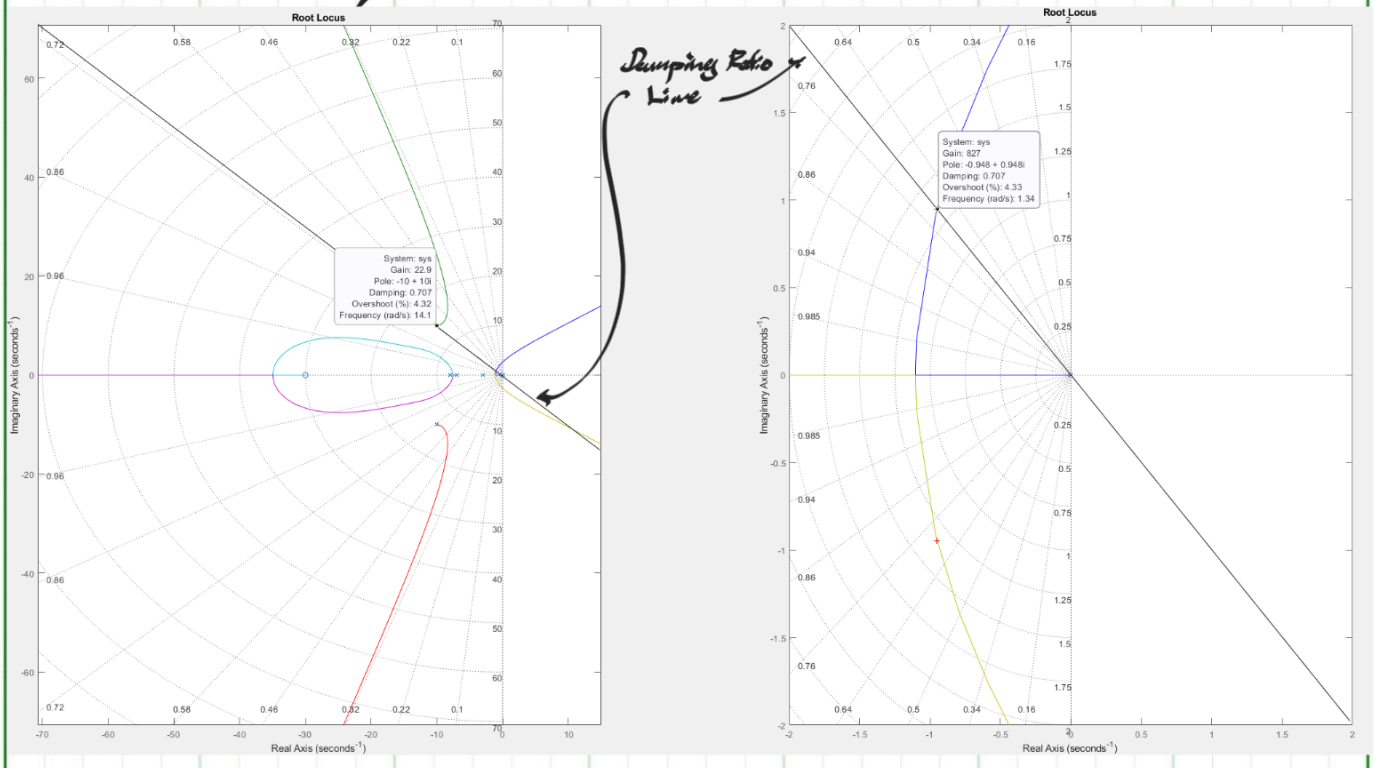
Name	Numerator	Denominator	Parameter Tunability	Absolute Tolerance	Continuous State Attributes
Transfer Fcn	5	1	Auto	auto	''
Transfer Fcn1	7	[1 2]	Auto	auto	''
Transfer Fcn10	7	[1 2]	Auto	auto	''
Transfer Fcn11	1	5	Auto	auto	''
Transfer Fcn2	7	[1 2]	Auto	auto	''
Transfer Fcn3	1	5	Auto	auto	''
Transfer Fcn4	5	1	Auto	auto	''
Transfer Fcn5	5	1	Auto	auto	''
Transfer Fcn6	7	[1 2]	Auto	auto	''
Transfer Fcn7	5	1	Auto	auto	''
Transfer Fcn8	7	[1 2]	Auto	auto	''
Transfer Fcn9	5	1	Auto	auto	''

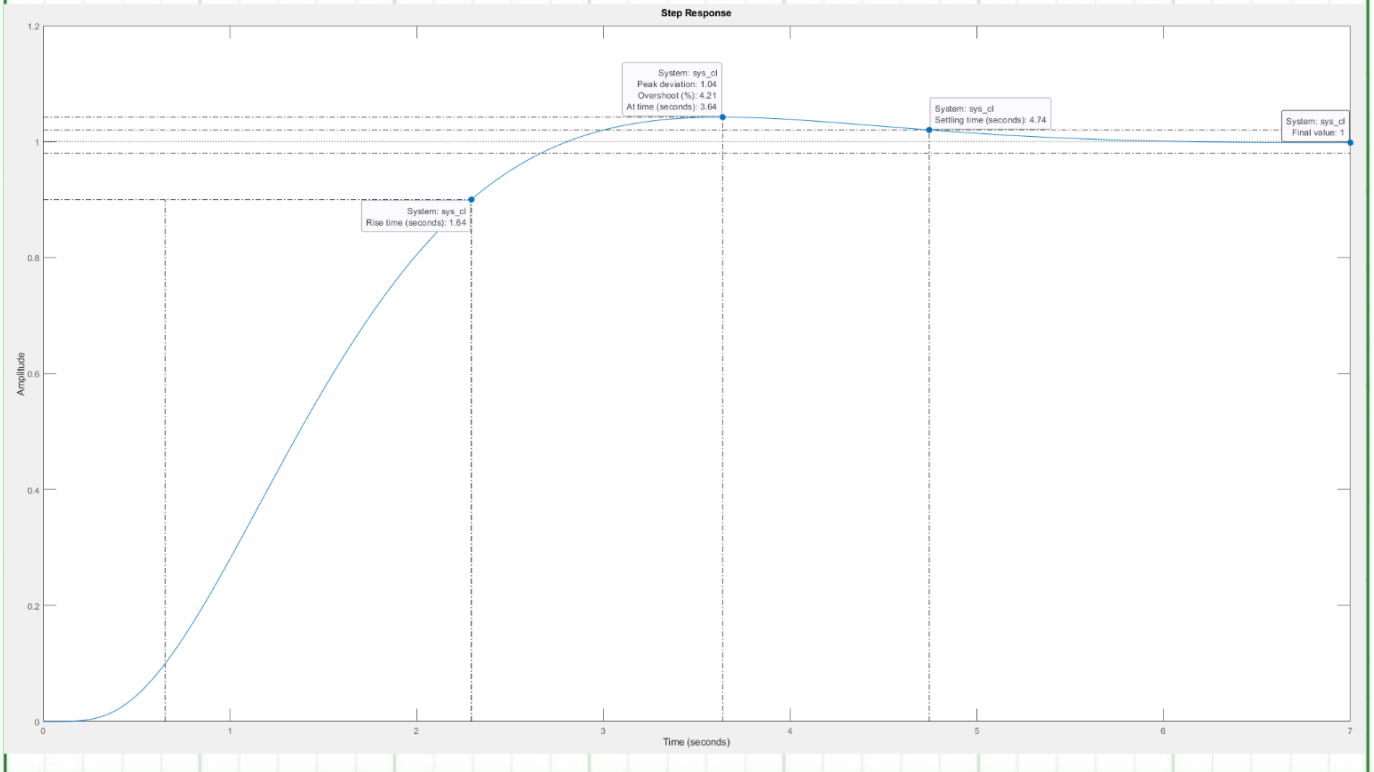
Appendix

Table 6. Block Type Count

BlockType	Count	Block Names
TransferFcn	12	Transfer Fcn, Transfer Fcn1, Transfer Fcn10, Transfer Fcn11, Transfer Fcn2, Transfer Fcn3, Transfer Fcn4, Transfer Fcn5, Transfer Fcn6, Transfer Fcn7, Transfer Fcn8, Transfer Fcn9
ToWorkspace	12	To Workspace, To Workspace1, To Workspace10, To Workspace11, To Workspace2, To Workspace3, To Workspace4, To Workspace5, To Workspace6, To Workspace7, To Workspace8, To Workspace9
Step	9	Step, Step1, Step2, Step3, Step4, Step5, Step6, Step7, Step8
Sum	8	Sum, Sum1, Sum2, Sum3, Sum4, Sum5, Sum6, Sum7
Scope	7	Scope, Scope1, Scope2, Scope3, Scope4, Scope5, Scope6
Constant	1	Constant1

Appendix B


Root Locus
System: sys
Gain: 22.9
Pole: -10 + 10i
Damping: 0.707
Overshoot (%): 4.32
Frequency (rad/s): 14.1
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
Damping Ratio Line
Root Locus
System: sys
Gain: 827
Pole: -0.948 + 0.948i
Damping: 0.707
Overshoot (%): 4.33
Frequency (rad/s): 1.34
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})


Step Response
System: sys_cl
Peak deviation: 1.04
Overshoot (%): 4.21
At time (seconds): 3.64
System: sys_cl
Settling time (seconds): 4.74
System: sys_cl
Final value: 1
System: sys_cl
Rise time (seconds): 1.64
Amplitude
Time (seconds)

```
clear
clc
clf
%%%---Run One section at a time---%%%

%set up transfer function
s=tf('s');
sys=(s+30)/(s*(s+3)*(s+7)*(s+8)*(s^(2)+20*s+200));

%set up axis system limits with the first indexing set so it makes
%overlaying the damping ratio line easier. set the second index to the
%zoomed in perameters for the root locus

axis_lim=[[-70.7,15,-70.7,70.7];[-2,2,-2,2]];

% set the perscribed damping ratio

Damp_ratio=0.707;

%Plot the root locus in subplots using rlocus() with an sgrid() overlay w/
%respect to the dampio ratio. this grid w/ an overlayed(manually) damping
%ratio will help in determining real values of gain that would stablize the
%sytem into an underdamped transform.

hold on
figure(1)
for i=1:2
    subplot(1,2,i)
    sgrid(Damp_ratio)
    rlocus(sys)
    axis(axis_lim(i,:))
j=i;
end
%%
%%%--- Run section---%%%

%Using Rlocfind lets you manually anaylize point locations on the root
%locus, pick location were the damping ratio overlayed line crosses with
%the root locus on the zoomed plot. k= gain
[k ,poles]=rlocfind(sys)
%%
%%%---Run Section---%%%

%use feedback(T(s),H(s)) function to compile the (sys*k)=T(s) with H(s)=1
% into the desired closed loop. use step()function to produce the stable underdamped
% step response.
sys_cl=feedback(k*sys,1)
figure(2)
step(sys_cl)
```

```
Select a point in the graphics window

selected_point =

  -0.9467 + 0.9515i


k =

  828.8086


poles =

  -9.9500 +10.0085i
  -9.9500 -10.0085i
  -8.1011 + 1.8585i
  -8.1011 - 1.8585i
  -0.9489 + 0.9522i
  -0.9489 - 0.9522i


sys_cl =

                          828.8 s + 2.486e04
  ------------------------------------------------------------------------
  s^6 + 38 s^5 + 661 s^4 + 5788 s^3 + 23560 s^2 + 3.443e04 s + 2.486e04

Continuous-time transfer function.
Model Properties
>>
```